

JAKKI CHAITANYA ESWAR RAJESH

AI Engineer | Azure Data Engineer | Python Developer

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LinkedIn | GitHub: github.com/ChaitanyaEswarRajeshJakki | AppNova AI | AriesGPT

■ PROFESSIONAL SUMMARY

Microsoft Certified Azure Data Engineer Associate (DP-203) with 4 years 10 months of hands-on IT experience building production Generative AI systems, voice AI pipelines, and cloud-scale ETL pipelines. Currently leading development of AppNova AI — an enterprise-grade, full-stack multi-agent LLM platform with FastAPI, React/TypeScript, LangGraph, ChromaDB RAG, and 5 LLM providers (Claude, Gemini, Groq, Ollama). Also built AriesGPT — a 5-microservice AI platform including a production VoiceBot (Whisper STT + Groq LLaMA 3 + gTTS TTS) with async FastAPI endpoints and structured JSON outputs. Proven track record architecting 30+ Azure Data Factory ETL pipelines for U.S. government agencies, reducing data processing time by 50%. Available for remote or Hyderabad-based AI Engineer and Azure Data Engineer roles.

■ TECHNICAL SKILLS

AI & LLMs: LangChain, LangGraph, Multi-Agent Orchestration, RAG, Prompt Engineering (CoT, ReAct, Few-Shot), Claude, Gemini, Groq (LLaMA 3), Ollama, Whisper (STT), gTTS (TTS), Edge TTS, Voice Pipeline Design, ViT-GPT2, Hugging Face Transformers, Sentence Transformers

Vector Databases: ChromaDB, Semantic Search, Embedding Pipelines, Corpus Deduplication, Content Fingerprinting

Cloud & Azure: Azure Data Factory (ADF), Azure Data Lake Gen2, Azure SQL Database, Azure Key Vault, Azure VMs, Azure Elastic Pools, Azure Logic Apps, Azure Storage

Programming: Python (Advanced), FastAPI, Flask, asyncio, Pydantic, SQL, Bash, REST APIs, Structured JSON Outputs, Confidence Thresholds, Escalation Logic

Data Engineering: ETL Pipeline Design, ADF Mapping Data Flows, Cloud Migration (On-Prem → Azure), Data Warehousing, Data Validation & Transformation, Data Modeling

AI/ML Libraries: OpenCV, FFmpeg, MoviePy, Pillow, scikit-learn, pandas, NumPy, PyTorch (inference), python-docx, PyMuPDF, gTTS, GFPGAN, DeblurGANv2

DevOps & Tools: GitHub Actions (CI/CD), Docker (basic), SSE Streaming, Loguru, Git, React (TypeScript/Vite), Angular (frontend integration), Uvicorn

Certifications: Azure Data Engineer Associate — DP-203 (Microsoft)

■ PROFESSIONAL EXPERIENCE

AI Engineer & Azure Data Engineer | West Advanced Technologies India Pvt. Ltd. | Hyderabad

Aug 2021 – Present | IT Experience: 4 years 10 months

AI & GenAI Application Development

- Built AppNova AI — an enterprise full-stack multi-agent platform using FastAPI (backend) + React/TypeScript/Vite (frontend), with 12 specialized LangGraph agents (architecture, code-analysis, code-generation, business-rules, security, DevOps, documentation, testing, data-migration, UI/UX, integration, migration-planner) that analyse legacy codebases and generate structured modernisation reports as DOCX/PDF exports.
- Implemented a 5-provider LLM cascade (Claude Sonnet, Gemini Flash, Groq LLaMA 3, Ollama) with per-key round-robin rotation across 25+ API keys, rate-limit detection, automatic provider fallback, and content-fingerprint caching — achieving near-zero idle time across all concurrent agents.
- Designed ChromaDB-backed RAG pipeline with corpus deduplication, AST-based file summaries, and semantic search — allowing agents to retrieve code-relevant context in under 200ms regardless of project size.
- Built AriesGPT — an Angular + FastAPI AI monorepo with 5 independent microservices: VoiceBot (Whisper STT + Groq LLaMA 3 + gTTS TTS), Video to Narrative (OpenCV + ViT-GPT2 + Whisper + Groq), Facial Deblur (GFPGAN), Vehicle Deblur (DeblurGANv2), and Aries GPT conversational assistant — all integrated into a single Angular dashboard.
- VoiceBot pipeline: browser audio input → Whisper speech-to-text transcription → Groq LLaMA 3 intent understanding and response generation with structured JSON outputs → gTTS text-to-speech synthesis returned to frontend; built with async FastAPI endpoints, strict request/response schemas, and escalation logic for failed transcription or LLM timeouts.
- Deployed GovGenie — a RAG-powered RFP response generator (Ollama + LangChain + ChromaDB) that retrieves past proposal language semantically and auto-populates government RFP templates, reducing bid writing time by ~70%.

- Built Reference Filler — a FastAPI + ChromaDB document automation system that parses uploaded resumes with LLM, extracts structured data, and fills DOCX templates — replacing a manual 45-minute HR task with a 2-minute automated workflow.
- Implemented CI/CD pipelines using GitHub Actions for zero-downtime automated deployment of all AI applications.

Azure Data Factory & ETL Pipeline Engineering

- Architected 30+ production ADF pipelines for U.S. government clients (Contra Costa County, California State Agencies, LA County) ingesting high-volume RMS records into Azure SQL Database and Data Lake with zero data loss.
- Designed end-to-end ETL workflows using Mapping Data Flows, Lookup activities, and parameterized Linked Services — achieving 50% faster data processing versus the prior manual batch process.
- Automated validation, deduplication, and transformation logic in ADF, cutting manual data review effort by 40% and providing real-time pipeline health visibility via Azure Monitor alerts.
- Led ARIES cloud migration — moved a large-scale criminal justice system (40+ tables, multi-million row datasets) from on-premises SQL Server to Azure Cloud with zero data loss and a full lineage audit trail.
- Maintained Linked Services, Integration Runtimes, and self-hosted IR configurations across Azure SQL, SFTP, REST, and Blob sources with secure credential management via Azure Key Vault.

■ CAREER TRANSITION & PRIOR WORK HISTORY

The following section documents my background before entering the IT industry in 2021. These roles are included for transparency and completeness. They are non-IT positions and are not relevant to my technical capabilities.

Career Transition & Self-Directed Learning | Dec 2018 – Jul 2021 (2 years 8 months)

- Completed self-directed study in Python, SQL, and cloud data engineering fundamentals including the COVID-19 pandemic period (2020–2021).
- Earned Microsoft Azure Data Engineer Associate (DP-203) certification — foundation for joining WATI in August 2021.

Office Co-Ordinator | Eastern Petro Technologies India Pvt. Ltd. | Kakinada | Apr 2018 – Nov 2018

Non-IT role — included for employment history completeness | EPFO verified

- Managed quotations, purchase orders, and cross-departmental coordination reporting to the CEO.

Process Associate | Macray Technologies Pvt. Ltd. | Kakinada | Oct 2016 – Mar 2017

Non-IT role — earliest career position | Small unregistered firm, no EPFO

- U.S. solar energy market cold calling and customer acquisition for Home Solar Solutions, West Palm Beach, Florida.

■ KEY PROJECTS

VoiceBot — End-to-End Voice Assistant Microservice | FastAPI · Whisper · Groq LLaMA 3 · gTTS · Angular | 2025

- Built production voice assistant as part of AriesGPT monorepo: speech captured via browser → Whisper STT transcription → Groq LLaMA 3 response generation with structured JSON output → gTTS voice synthesis returned to Angular frontend.
- Implemented async FastAPI endpoints with strict request/response schemas, confidence threshold logic, and escalation handling for failed transcription or LLM timeouts — designed for production reliability.
- Pipeline built for extensibility: fallback responses, error state management, and per-request logging via rotating file handlers.

AppNova AI — Enterprise Multi-Agent Code Modernisation Platform | FastAPI · React/TypeScript/Vite · LangGraph · Claude · Gemini · Groq · ChromaDB · asyncio · SSE | 2025 – Present

- Full-stack AI platform with 13 Claude-driven LangGraph agents for legacy code modernization, producing runnable projects, audits, and .md/.docx/.pdf reports.
- Engineered 5-provider LLM cascade with automatic key rotation across 25+ API keys — agents never stall on rate-limits.
- SHA-256 content-fingerprint caching lets repeated uploads skip all LLM calls entirely, serving ChromaDB results in seconds.
- SSE streaming with fan-out queue architecture, real-time agent thinking logs, per-agent cancellation, and Spectator Mode.

AriesGPT — Angular + FastAPI AI Monorepo | Angular · FastAPI · GFPGAN · DeblurGANv2 · Whisper · ViT-GPT2 · Groq | 2025

- Unified Angular + FastAPI workspace with 5 independent AI microservices: VoiceBot, Video to Narrative, Aries GPT assistant, GFPGAN facial deblur, and DeblurGANv2 vehicle deblur.
- Each service runs independently with its own venv and requirements; CORS handled via environment-based origin configuration.

Video to Narrative — Surveillance Video Summarisation App | Flask · OpenCV · Whisper · ViT-GPT2 · Groq (LLaMA 3) | Completed 2025

- Flask AI app: extracts frames with OpenCV at configurable intervals, generates per-frame captions via ViT-GPT2, transcribes audio with Whisper, then uses Groq LLaMA 3 to produce both plain-English and formal law-enforcement incident summaries as structured JSON.
- GPU/CPU auto-detection via PyTorch; automatic cleanup of frames and uploaded video post-processing.

GovGenie — AI-Powered RFP Response Generator | FastAPI · Ollama (LLaMA 3.1) · LangChain · ChromaDB · Hugging Face | 2025

- RAG system that reads an uploaded government RFP PDF, retrieves semantically similar past proposal sections from ChromaDB, and drafts section-by-section responses — reducing RFP writing effort by ~70%.
- ConversationBufferMemory for follow-up clause questions; per-stage token metrics via tiktoken for cost visibility.

ReferenceFiller — Automated DOCX Template Population | FastAPI · ChromaDB · Ollama · python-docx · Jinja2 | 2025

- HR uploads resume + template → LLM extracts structured candidate data → fills DOCX via semantic field mapping — turning 45-minute manual task into 2-minute automated workflow.

Gemini YouTube Automation Bot — Autonomous AI Video Publisher | Python · Gemini · MoviePy · gTTS · GitHub Actions · YouTube Data API | Personal

- Fully autonomous pipeline: Gemini generates SEO-optimised scripts → gTTS voiceover → MoviePy video composition → YouTube Data API auto-publish — entire workflow runs on GitHub Actions cron, no server required.

AI Content Bot — Fully Automated Daily AI/Tech Video Pipeline | Python · Gemini 2.5 Flash · Microsoft Edge TTS · MoviePy · FFmpeg · Pexels API · Cloudinary · YouTube Data API · Instagram Graph API · GitHub Actions | 2025 – Present

- Fully automated daily pipeline: fetches trending AI & Tech news from RSS feeds (TechCrunch, VentureBeat, The Hacker News, AI News) → Gemini 2.5 Flash generates a 45-second script, Instagram caption, 20 hashtags, and a topic-matched Pexels search keyword → Microsoft Edge TTS generates voiceover with word-level timing for animated captions → MoviePy + FFmpeg assembles 3 Pexels stock clips into a 1080×1920 vertical video with animated word-synced captions and title overlay.
- Publishes automatically to YouTube Shorts (YouTube Data API v3) and Instagram Reels (Cloudinary CDN + Instagram Graph API) — 100% hands-free via GitHub Actions daily cron at 6:00 AM IST.
- Built a Gemini 2.5 Flash auto-fix agent: on pipeline failure, downloads the job log, analyses it alongside all source files, and opens a pull request with patched code automatically — no manual intervention required.

AppNovaAI — AI-driven code modernization studio — drives Claude Code headless through a DAG of 14 specialist agents to convert legacy codebases into runnable target stacks. | architecture-diagrams-generation, claude-max, code-generation-ai-agents, dag-scheduler, developer-tools, docx-export, llm-orchestration-agent-ai, mermaid-diagrams, pdf-export, security-auditing-tool, self-hosted, self-hosted-ai, self-hosted-runner, sse-streaming, static-analysis | 2025

- AppNovaAI modernizes legacy codebases via Claude agents, providing runnable projects with build manifests, architecture diagrams, security audits, blueprints, and reports.
- Implemented advanced features in AppNovaAI including auto-generated build manifests and parallel chunk code generation, while resolving critical UI and pipeline stability issues.

AppNova_Working_09-04-2026 — AI-powered codebase intelligence platform: chat with your repo, generate architecture diagrams, migrate legacy code, convert files to modern stacks, and export reports — powered by Claude Code Max; local-first, privacy-safe, all major languages supported. | ai, architecture-diagrams, chromadb, code-exploration, code-generation, codebase-intelligence, conversion-v2-python, dag, fastapi, langchain, llm, migration, react | 2025

- Developed an AI-powered codebase intelligence platform (Python, FastAPI, React) with Claude AI CLI integration for querying, explanation, and migration plans.
- Re-architected the platform by integrating a Claude Code CLI runner, replacing the LangGraph and multi-LLM pipeline for advanced code analysis.
- Implemented an auto-sync CI workflow for documentation and integrated `appnova\_claude-cli` features with the current React frontend.

AriesGPT — AriesGPT is a unified Angular + FastAPI suite for AI media workflows: facial and vehicle deblurring (GFPGAN, DeblurGANv2), video to narrative, a GPT-powered assistant, and a VoiceBot UI. Env-based configs and git submodules make local setup fast, secure, and ready for experimentation and collaboration .Fast, secure local prototyping and integration | angular, cors, deblurganv2, environmentvariables, fastapi, gfpgan, git, gpt, httpclient, monorepo, nlp, python, routing, scss, speectotext, submodules, texttospeech, typescript, videotonarrative, voicebot | 2025

- Developed AriesGPT, an AI-powered Angular+FastAPI monorepo for law enforcement and investigation with voice, NL→SQL, mapping, deblur, crime intelligence, and narrative.
- Implemented specialized AI modules for facial/vehicle deblur (GFPGAN, DeblurGANv2) and AI-generated incident narratives from frame-by-frame video captioning.
- Engineered a NL→SQL query engine over criminal records using Groq and ChromaDB, and a criminal intelligence database via Claude API.

GovGenie — AI-powered RFP response generator and resume-to-template processor for government and enterprise agencies. Built with Angular, FastAPI, LangChain, ChromaDB, and local Ollama LLMs. | ai, angular, chromadb, fastapi, langchain, local-llm, ollama, rag, resume-parser, rfp-generator | 2025

- Full-stack AI platform automating RFP response generation and resume population with local LLMs (no API costs) and RAG for experience.
- Engineered a Skill Matrix Filler leveraging LLMs to auto-populate resume data into DOCX templates, generating contextual content for placeholders.
- Built a full-stack AI platform, GovGenie, automating RFP responses and resume population using a local LLM (Ollama), eliminating external API costs.

ModernizeAI — ModernizeAI is a full-stack AI-assisted platform to modernize workflows and applications with a Python backend, TypeScript frontend, and intelligent automation. | angular, batchfile, bootstrap, css, fastapi, html, numpy, pandas, python, react, scikit-learn, typescript | 2025

- Developed a full-stack AI-assisted generator for multi-stack apps, data, and validation, using Angular 18, FastAPI, Streamlit, Ollama, and downloadable packaging.
- Designed an Angular UI to configure application generation, prompting FastAPI to produce stack-specific artifacts and package them into a downloadable ZIP.

Skill_Matrix_Streamlit_App — AI-powered skill matrix filler using Streamlit, FastAPI, ChromaDB RAG, and local Ollama LLM — no cloud API keys needed. | chromadb, document-automation, fastapi, llm, local-llm, ollama, python, rag, resume, skill-matrix, streamlit | 2025

- Developed an AI-powered Streamlit/FastAPI app using local Ollama and ChromaDB to parse resumes, fill templates, and provide RAG Q&A;
- Engineered robust resume parsing from DOCX/PDF via sequential prompting, integrated ChromaDB for RAG Q&A, and enabled browser preview and download of filled documents.
- Implemented session isolation with unique IDs and dedicated ChromaDB collections per upload, ensuring data privacy and preventing cross-session data leakage.

AppNovaAI — AI-driven code modernization studio — drives Claude Code headless through a DAG of 14 specialist agents to convert legacy codebases into runnable target stacks. | architecture-diagrams-generation, claude-max, code-generation-ai-agents, dag-scheduler, developer-tools, docx-export, llm-orchestration-agentic-ai, mermaid-diagrams, pdf-export, security-auditing-tool, self-hosted, self-hosted-ai, self-hosted-runner, sse-streaming, static-analysis | 2025

- AppNovaAI modernizes legacy code into runnable, audited stacks using Claude Code, DAG-driven RAG agents, guardrails, and interactive retest loops.
- Engineered AppNovaAI to automatically generate architecture diagrams, security audits, and migration blueprints, producing comprehensive .md, .docx, and .pdf reports.
- Optimized AppNovaAI's codegen pipeline with parallel chunk waves and deterministic file_map backfill, while enhancing UI and automated build manifest generation.

ModernizeAI — ModernizeAI is a full-stack AI-assisted platform to modernize workflows and applications with a Python backend, TypeScript frontend, and intelligent automation. | angular, batchfile, bootstrap, css, fastapi, html, numpy, pandas, python, react, scikit-learn, typescript | 2025

- A local full-stack AI-assisted starter generation tool with Angular 18, FastAPI, Ollama, and optional Streamlit, packaging diverse stacks into ZIPs.
- Implemented backend logic to generate tabular data, UI, middle-tier, and backend artifacts, packaging all outputs into a downloadable ZIP based on user selections.
- Enabled users via an Angular UI to select preferred technology stacks (UI, middle-tier, backend) for AI-assisted code generation and automated deployment packaging.

Skill_Matrix_Streamlit_App — AI-powered skill matrix filler using Streamlit, FastAPI, ChromaDB RAG, and local Ollama LLM — no cloud API keys needed. | chromadb, document-automation, fastapi, llm, local-llm, ollama, python, rag, resume, skill-matrix, streamlit | 2025

- AI-powered Streamlit app using FastAPI, ChromaDB, and local Ollama LLM to parse resumes, fill templates, and answer questions via RAG.
- Implemented RAG Q&A; functionality using ChromaDB for vector search, enabling natural-language queries grounded in document content; ensured secure session isolation.
- Engineered a full-stack architecture with a Streamlit frontend and FastAPI backend, leveraging a local Ollama LLM to eliminate cloud API dependencies.

instagram-god-bot — Automated bot that posts daily Bhagavad Gita verses as Instagram Reels in Hindi, Telugu & English using Gemini AI, Google TTS, Pexels, and FFmpeg | automation, bhagavad-gita, cloudinary, content-automation, daily-posts, ffmpeg, gemini-ai, github-actions, hindi, instagram-bot, krishna, multilingual, python, spiritual-content, spiritual-instagram-reels, telugu | 2025

- Automated GitHub Actions bot publishes daily Bhagavad Gita verses in Hindi, Telugu, English as Reels/Shorts using Gemini AI, Google TTS, Pillow, FFmpeg, Cloudinary.
- Engineered a multilingual video generation pipeline for Instagram Reels, integrating Pillow for pixel-accurate text overlay and FFmpeg for H.264 video composition, ensuring Indic script rendering.
- Designed a hands-free automated pipeline to post daily Instagram Reels, utilizing Gemini AI for content, Google TTS for voiceovers, and Instagram Graph API for publishing.

■ EDUCATION & CERTIFICATIONS

- B.Tech, Mechanical Engineering | GIET College of Engineering, Rajahmundry | May 2015
- Microsoft Certified: Azure Data Engineer Associate (DP-203) | Microsoft

■ GOVERNMENT CLIENT ENGAGEMENTS (VIA WEST ADVANCED TECHNOLOGIES)

- **Contra Costa County, California** — ADF pipelines for RMS data migration; 30+ entity tables synced between legacy on-prem SQL Server and Azure SQL Database.
- **California State Agencies** — Parameterized ADF pipelines for real-time reporting and compliance data supporting multiple agencies from a single pipeline template.
- **Los Angeles County** — BEAGLE system optimization and ARIES cloud migration; delivered improved query performance, compliance audit trails, and documented DR/rollback procedures.
- **WATI (Government Contractor)** — GovGenie RFP automation platform; reduced proposal preparation from 5+ days to under 24 hours for standard government bid responses.